

BRAINS BEHIND THE TECH
Sreetosh Goswami (seated) and Prof. Navakanta Bhat at their lab in IISc, Bengaluru

HEMANT MISHRA

roundtrips that data packets would have to traverse. Besides the savings on energy, it also means that critical data—like in the health sector where privacy is essential or in strategic defence applications—can be kept secure.

Neuromorphic design and its pursuit of brain-like processing is still an emerging field, one in which top technology companies such as the US-based Intel and IBM—besides a slew of specialist start-ups incubated at university level—are invested in, either in terms of better chip design or in algorithms or software. In terms of hardware, researchers globally have been experimenting with a variety of devices such as memristors, ferroelectric devices and resistive RAM. For instance, one of the main challenges in memristors has been to achieve a large number of unique states in which the device could store data. It's this challenge that the IISc device has overcome.

Enter the Matrix

Think of this device as a grid of memristors, storing data in between the junctions but in 16,500 different states of conductance. When a voltage is applied through the rows and columns of this grid, it interacts with these various states of conductance. This grid pattern also represents how some basic multiplication takes place in a computer—called matrix multiplication. In a traditional computer, matrix multiplication of a grid of 100x100 will require 10,000 steps. But the analogue accelerator requires just one step to get to the answer because the data is readily held in a spectrum of various states at the junctions of the matrix. Moreover, the research paper demonstrated that the accelerator's energy efficiency was 250 times that of a GPU (of the American tech giant Nvidia) during a matrix multiplication task.

"This geometric structure itself does the matrix multiplication in one step," explains Goswami who began working on this problem back in 2011 when he was an electrical engineering student at the Indian Institute of Engineering Science and Technology in Shibpur, Howrah. Inspiration came from his father, Prof. Sreebrata Goswami, an inorganic chemistry expert who was researching the properties of various molecules in new chemical combinations. "That's how our paths crossed... academically," he quips. "I got interested in turning these into devices."

SEVERAL STEPS REMAIN IN TRANSFORMING THE IISc MEMRISTOR FROM A LAB-SCALE DEVICE TO A FINAL PRODUCT READY FOR MARKET

After a stint at the National University of Singapore for his PhD, Goswami headed to Bengaluru to join the IISc where he continued research on the accelerator. While his father worked on the design of the molecular system, Goswami teamed up with Prof. Bhat and four research fellows—Deepak Sharma, Santi Prasad Rath, Bidyabhusan Kundu and Harivignesh S. The quartet tackled the circuit and system design, electrical characterisation, fabrication, mathematical modelling and the bio-inspired neuronal response behaviour.

"This is clearly a breakthrough," says Gopalakrishnan who, apart from running a technology seed fund called Axilor Ventures, is also the IISc Council chairman. However, there are several steps involved in transforming the memristor from a lab-scale device to a final product ready for market, he adds. This will involve miniaturising the device, writing the software algorithms that will allow it to communicate with a digital chip and, finally, manufacturing it. "If you are able to do all that," he adds, "this is a significant breakthrough and India has something which should be taken to a logical conclusion by creating a business and an industry around it."

Prof. Bhat, an expert in silicon electronics, agrees there is a long way to go. Work on integrating the various parts of the memristor onto a chip is underway, he says. "In fact, we are working on the system-on-a-chip (SoC) right now. The first version will be taped out (that is, sent to a fabrication facility) by December this year. By first quarter next year, we will have some early results to share," says Bhat. Brewing simultaneously are plans for a start-up. The standout factor of the IISc memristor is how it evolved in classic fashion—from experiments in basic science leading up to a technological breakthrough. At the CeNSE lab, the excitement is palpable. ■